DPEN021: Enabling Mathematics A
Challenge Questions Week 2

Week 2B

Key Idea – Rules for Surds

1. Given the following question: move to homework

$$\sqrt{x^2+y^2} = \sqrt{x^2}+\sqrt{y^2}$$
$$= x+y$$

Are the following answers, correct? For each incorrect answer explain what the error is.

Answer1: The answer is correct because $\sqrt{4^2+4^2} = 2+2 = 4$

Answer2: The answer is correct because $\sqrt{x^2} = x$ and $\sqrt{y^2} = y$

Answer3: The error is in the first line. It is not correct to write $\sqrt{x^2+y^2} = \sqrt{x^2}+\sqrt{y^2}$

2. Find the square root of $16+2\sqrt{55}$.
3. Find the values of the pronumerals.

(a) $6-\sqrt{2} = a-\sqrt{b}$ (b) $x+\sqrt{y} = 3+\sqrt{10}$ (c) $2a+3\sqrt{b} = 4+3\sqrt{2}$

(d) $a\sqrt{3} = \sqrt{27}$ (e) $\sqrt{21}+3 = \sqrt{3}\left(\sqrt{a}+\sqrt{b}\right)$ (f) $(\sqrt{2}+\sqrt{5})^2 = a+b\sqrt{10}$

(g) $\left(\sqrt{5}-\sqrt{3}\right)\left(\sqrt{5}+\sqrt{3}\right) = c$ (h) $\sqrt{125}-\sqrt{x} = 2\sqrt{5}$ (i) $x+y\sqrt{3} = \left(2\sqrt{3}-1\right)\left(\sqrt{3}+2\right)$

Key Idea - Rationalising the Denominator

1. Express with a rational denominator:

(a) $\dfrac{\sqrt{2}-\sqrt{7}}{\sqrt{2}+\sqrt{3}}+\dfrac{3}{3\sqrt{2}-\sqrt{5}}$ (b) $\dfrac{1}{\sqrt{5}+\sqrt{2}}+\dfrac{3}{3\sqrt{2}-\sqrt{5}}$

2. Evaluate $\dfrac{3\sqrt{2}+4}{\sqrt{6}-\sqrt{3}}+\dfrac{\sqrt{2}-1}{\sqrt{6}-\sqrt{3}}-\dfrac{2}{\sqrt{6}-1}$ leaving your answer with rational denominator.

3. If $x = \dfrac{2+\sqrt{3}}{2-\sqrt{3}}$, evaluate $\dfrac{1}{x}+x$.

Week 2C

Key Idea - Combined Index Rules

1. Simplify 4^x+4^x
2. If $2^x = \dfrac{5}{2}$ and $2^y = 3$ then 2^{x+y} equals?
3. Simplify $\left(x^2-y^2\right)^{\frac{1}{2}} \times \left(x-y\right)^{\frac{3}{2}} \times \left(x+y\right)^{\frac{-1}{2}}$
4. Simplify the following expressing with positive indices and as a single fraction where needed:

(a) $1+a^{-1}$ (b) $\dfrac{a^{-7}+12a^{-1}}{1-3a^{-1}}$ (c) $\left(a^{-3}+a^{-1}\right)^2$ (d) $\dfrac{x-5+6x^{-1}}{1-2x^{-1}}$

5. The intensity I of sound is given by $I = I_0 \times 10^{\left(\frac{L}{10}\right)}$, Where I_0 is the reference intensity and L is the sound level in decibels (dB). If the sound level L is increased by 20 dB, how does the intensity I change?

6. The amount A of a radioactive substance remaining after time t is given by $A = A_0 \times 2^{-\frac{t}{T}}$ where A_0 is the initial amount and T is the half-life. If the time t is tripled, how does the remaining amount A change?

7. An electrical engineer is analysing the impedance of a circuit component, which is given by $\left(\frac{27}{8}\right)^{\frac{2}{3}}$. Simplify this expression to find the impedance in a more manageable form.

8. A civil engineer is designing a cantilever beam with a deflection formula given by $d = \left(\frac{4}{9}\right)^{\frac{1}{2}} L$ where d is the deflection at the end of the beam and L is the length of the beam. Simplify this expression to find the deflection in terms of L.